

433MHz RF Modules

Cheap Wireless Remote Control

Core Concept

433MHz RF = Radio Frequency communication at 433.92 MHz
ISM Band = Industrial, Scientific, Medical (unlicensed spectrum)



Real analogy: Like a walkie-talkie that only sends beeps - very simple, but cheap and works through walls.

Common names:

- 433MHz RF modules
- ASK/OOK transmitter/receiver
- Superheterodyne receiver
- Remote control modules

Why 433MHz RF?

Advantages:

- **Extremely cheap:** \$1-3 complete TX/RX
- **No line-of-sight needed:** Works through walls
- **Simple to use:** Just digital HIGH/LOW
- **Good range:** 50-500m depending on antenna
- **Low power:** TX only during transmission
- **No pairing:** Just turn on and go
- **Unlicensed:** Free to use in most countries

Disadvantages:

- **Very slow:** Typically 1-10 kbps
- **No built-in protocol:** Must implement yourself
- **No encryption:** Security is DIY
- **Half-duplex:** Can't TX and RX simultaneously
- **Prone to interference:** Noisy frequency band
- **One-way typically:** RX modules can't send back

Perfect for: Garage doors, wireless switches, doorbells, simple remotes

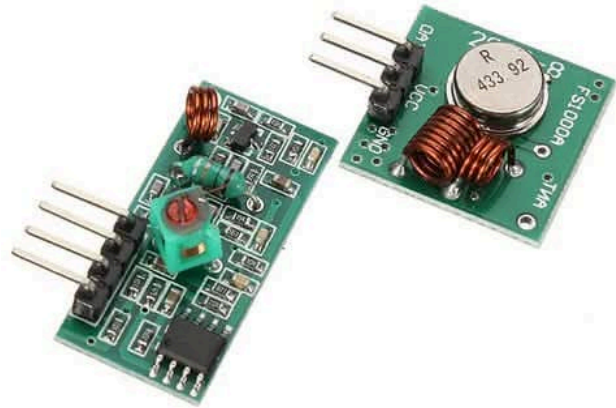


433MHz vs Other Technologies

Feature	433MHz RF	IR	Bluetooth	WiFi	LoRa
Cost	Very Low (\$1-3)	Very Low (\$1)	Low (\$2-5)	Low (\$2-5)	Low (\$5-15)
Range	100-500m	1-10m	10-100m	30-50m	2-15 km
Line-of-sight	No	Yes	No	No	No
Data Rate	1-10 kbps	1-5 kbps	1-3 Mbps	50+ Mbps	0.3-50 kbps
Power	Very Low	Very Low	Low	High	Very Low
Walls	Yes (good)	No	Yes (fair)	Yes (fair)	Yes (excellent)
Security	None	Physical	Pairing	WPA2/3	Optional
Complexity	Very Simple	Very Simple	Medium	High	Medium
Protocol	DIY	Fixed codes	Standard	Standard	Standard
Bidirectional	Hard	Possible	Yes	Yes	Yes

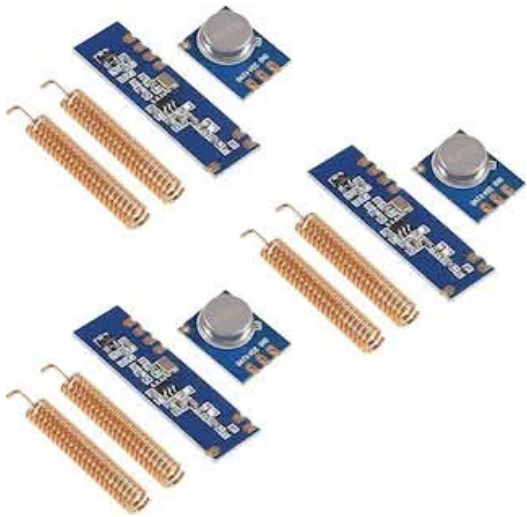
Common 433MHz Hardware Modules

FS1000A/XY-FST



- Frequency: 433.92 MHz
- Power: 10-40 mW (adjustable)
- Voltage: 3-12V
- Current: 10-40mA when transmitting
- Range: 20-100m (with poor antenna)
- Price: ~\$1
- 3 pins: VCC, GND, DATA

STX882 (Better quality)



- Frequency: 433.92 MHz
- Power: 25mW
- More stable than FS1000A
- Better sensitivity
- Price: ~\$2

Pin Connections (Arduino)

Transmitter (FS1000A):

FS1000A TX
VCC → 5V
GND → GND
DATA → D12

(Any digital pin)

- Use 5V for VCC (some modules can do 3.3V but 5V is more common)
- Without antenna, range is very poor (1-10m)

Receiver (XY-MK-5V):

XY-MK-5V RX

VCC → 5V

GND → GND

DATA → D2

DATA2 → NC

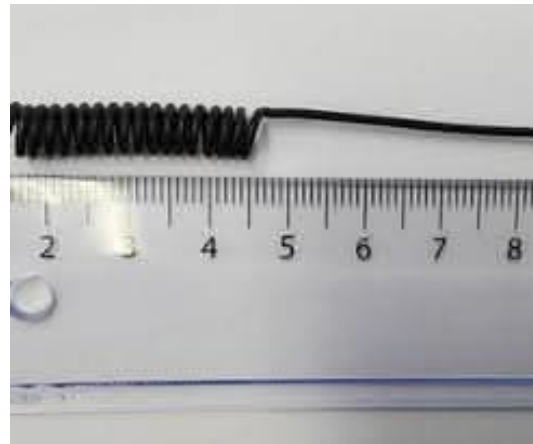
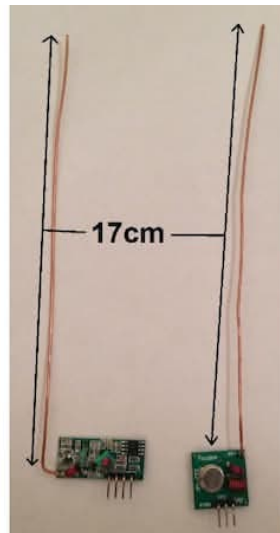
(Interrupt pin recommended)
(Not used)

- D2, D3, D18, D19 are interrupt pins on Arduino Uno
- Use interrupt for better performance (detect signal changes immediately)

Antenna:

- 17.3 cm wire (1/4 wavelength for 433MHz)
- Calculation: $299,792 / 433.92 / 4 = 17.3 \text{ cm}$
- Vertical orientation is best
- Longer = better range (up to 1/2 wave = 34.6 cm)

Critical: Antenna is REQUIRED for good range!



How 433MHz Modulation Works

ASK/OOK Modulation

ASK = Amplitude Shift Keying

OOK = On-Off Keying (simplest form of ASK)

Digital 1: Carrier ON (full power)

Digital 0: Carrier OFF (no power)

Example transmission of "1 0 1 1 0":

Carrier					
Power	[HIGH]	[LOW]	[HIGH]	[HIGH]	[LOW]
Bit	1	0	1	1	0

No fancy modulation like WiFi/Bluetooth!
Just turn transmitter on/off rapidly.

Why so simple?

- Very cheap to implement
- Receiver just detects carrier presence
- No complex demodulation needed

However, this simplicity comes with drawbacks:

- Not spectrum efficient
- Prone to interference
- No error correction

Data Encoding Methods

Raw Digital (Simplest - Don't Use!)

```
// BAD: Just send raw digital pulses
digitalWrite(TX_PIN, HIGH); // Transmit 1
delay(1);
digitalWrite(TX_PIN, LOW);  // Transmit 0
delay(1);
```

NRZ Encoding

- We call this encoding NRZ (Non-Return to Zero)

1111 → HHHH (High)
0000 → LLLL (Low)

Problems:

- No synchronization
- Receiver can't detect start
- Clock drift causes errors
- No error detection

Manchester Encoding (Better)

Manchester encoding: Edge = clock + data

Bit 0: High-to-Low transition

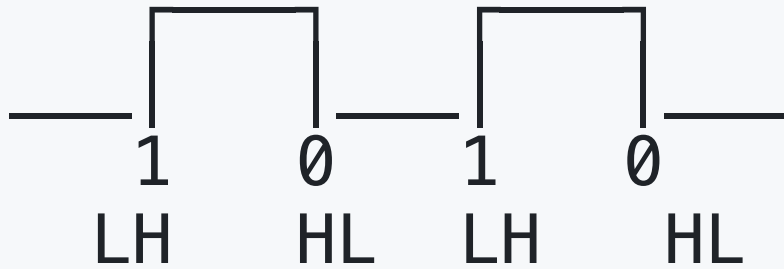


Bit 1: Low-to-High transition



- The transition (0 -> 1 or 1 -> 0) encodes the bit value
- To make the transitions, we need to send two pulses per bit (half the data rate)

Example "1010": "LHHLLHHL"



1111 → LH LH LH LH

0000 → HL HL HL HL

Benefits:

- Self-clocking (no clock drift)
- Always has transitions
- DC balanced

Disadvantages:

- Half the data rate (need 2 bits to encode 1)

Packet Structure (Recommended)

Example packet format:

Preamble 10101010	Sync 11110000	Length 1 byte	Payload N bytes	Address 1 byte	CRC 1 B
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Preamble: Wake up receiver, sync AGC

Sync: Unique pattern to detect packet start

Length: How many bytes follow

Payload: Your actual data

Address: Device ID (multiple devices on same frequency)

CRC: Error detection

Example packet (button press):

Preamble: 0xAA (10101010)

Sync: 0xF0 (11110000)

Length: 0x02 (2 bytes)

Payload: 0x01 0xFF (Button 1, ON)

Address: 0x42 (Device ID)

CRC: 0x3A (checksum)

Antenna Importance

Range = Antenna!

Wire Coil Spring Antenna (Cheapest):

- Cut wire to $1/4$ wavelength = 17.3 cm
- Calculation: $299,792 \text{ km/s} \div 433.92 \text{ MHz} \div 4$
- Solder directly to ANT pin
- Vertical orientation best
- Range: 50-200m

Whip Antenna:

- Rigid, adjustable
- Better impedance match
- 17.3 cm (1/4 wave) or 34.6 cm (1/2 wave)
- Range: 100-300m



Common 433MHz Problems & Solutions

Problem 1: Very Short Range (< 10m)

Symptoms:

- Works only when very close
- Unreliable at distance

Solutions:

1. Add proper antenna!

Most common problem: NO antenna

Solution: Add 17.3 cm wire

Result: 10x range improvement!

2. Check power supply

TX needs 10-40mA during transmission

Weak power = weak signal

Solution: Use good 5V supply, add capacitor

3. Optimize timing

```
// Too fast transmission  
mySwitch.setPulseLength(100); // x Too short!  
  
// Better  
mySwitch.setPulseLength(350); // - Standard
```

Problem 2: No Reception / Doesn't Decode

Symptoms:

- RX module seems dead
- Serial Monitor shows nothing

Solutions:

1. Check wiring

RX Module Pins:

VCC → 5V (check polarity!)

GND → GND

DATA → Arduino pin 2 (interrupt pin)

Test with multimeter:

VCC should be 5V

GND should be 0V

2. Test with known remote

Use commercial remote (garage door, RF outlet)
Should see SOMETHING when button pressed
If not → module or wiring problem

3. Receiver sensitivity

Upload simple test code:

```
void loop() {  
    int value = digitalRead(RX_PIN);  
    Serial.println(value);  
    delay(10);  
}
```

Should see noise: 0101010101...

Press remote: should see pattern change

Problem 3: Interference / False Triggers

Symptoms:

- Random codes received
- Works sometimes, not others

Solutions:

1. 433MHz is **CROWDED!**

Common sources:

- Wireless doorbells/Car key fobs/Weather stations
- Security sensors/Baby monitors/Tire pressure sensors
- Neighbors' devices!

Solution:

- Move away from other RF sources
- Use unique codes
- Implement proper packet validation

2. Add filtering in code

```
// Validate received code
if (receivedCode > 0 && receivedCode < 1000000) {
    // Reasonable code range
    if (receivedCode == EXPECTED_CODE) {
        // Take action
    }
}
```

3. Use acknowledgment

```
// Send code multiple times
for (int i = 0; i < 5; i++) {
    sendCode();
    delay(10);
}

// RX: Only accept if received 3+ times
```

Problem 4: Through-Wall Range Poor

Symptoms:

- Good range outdoors
- Poor range indoors

Causes & Solutions:

1. Construction materials

- Concrete/brick: 50-70% signal loss
- Wood frame: 10-20% signal loss
- Metal studs: 70-90% signal loss (VERY BAD)

Solutions:

- Add repeater in middle
- Use higher TX power
- Position antennas near windows
- Use directional antennas

2. Interference from electronics

WiFi routers, microwave ovens, LED lights can cause interference

Solution:

- Move away from electronics
- Use shielding
- Consider different frequency (315MHz)

Range Testing Results

Test Setup:

- FS1000A TX module (25mW)
- XY-MK-5V RX module
- Arduino Uno (both sides)
- RCSwitch library
- 17.3 cm wire antennas (vertical)

Results:

Open Field (Line of Sight):

- Good reception: 200-300m
- Weak reception: 300-400m
- No reception: > 400m

Suburban (Through houses):

- Good: 50-100m
- Weak: 100-150m
- No reception: > 150m

Indoor (Same building):

- Wood walls: 20-50m
- Concrete walls: 10-20m
- Metal obstacles: 5-10m

From Outside → Inside:

- Through window: 50-100m
- Through wood wall: 30-50m
- Through brick wall: 10-20m
- Through concrete: 5-10m

Power Consumption

Battery Life Calculations

Transmitter (FS1000A):

Idle: 0 mA (completely off when not transmitting!)

Transmitting: 20-40 mA (brief bursts)

Average (1 press/minute): ~0.1 mA

Battery life (2x AA = 2500 mAh):

$2500 \text{ mAh} \div 0.1 \text{ mA} = 25,000 \text{ hours} = 3 \text{ YEARS!}$

Receiver (XY-MK-5V):

Always on: 2-4 mA (listening)

Battery life (2x AA):

$2500 \text{ mAh} \div 3 \text{ mA} = 833 \text{ hours} = 35 \text{ days}$

Key insight:

- TX is VERY power efficient (only on when transmitting)
- RX drains battery (always listening)
- For battery operation: TX good, RX needs frequent charging

Solutions for RX:

- Use power adapter (not battery)
- Add solar panel
- Use sleep mode (wake on interrupt)

Using ISM Bands in the United States

For IoT, RF Modules, LoRa, and Wireless Projects

What is ISM?

ISM = Industrial, Scientific, and Medical

Originally reserved for:

- Industrial equipment
- Scientific instruments
- Medical devices (e.g., microwave ovens)

Now widely used for:

- Wi-Fi
- Bluetooth
- LoRa
- Remote controls
- IoT devices

No individual license required (if rules are followed).

Major ISM Bands in the U.S.

Band	Frequency Range	Common Use
5.8 GHz	5725–5850 MHz	Wi-Fi, video
2.4 GHz	2400–2483.5 MHz	Wi-Fi, Bluetooth
902–928 MHz	“915 MHz” band	LoRa, IoT
433 MHz	Limited use	Some remotes
315 MHz	Very common	Car key fobs

915 MHz — The Primary U.S. IoT Band

902–928 MHz

Why it's popular:

- Higher allowed power
- Longer range than 2.4 GHz
- Strong LoRa ecosystem
- Less congestion than 2.4 GHz

Most U.S. LoRa modules use this band.

433 MHz in the U.S.

- Allowed under FCC Part 15
- Not the primary ISM band
- Less common for new IoT designs
- Often replaced by 915 MHz

Important:

A module legal in Europe (433 MHz) may not be ideal for U.S. deployment.

315 MHz vs 433 MHz (U.S. Context)

Feature	315 MHz	433 MHz
Car remotes	Very common	Rare
IoT ecosystem	Weak	Weak
LoRa support	No	Rare
Module availability	Moderate	Moderate
U.S. focus	Yes	Not primary

In practice:

- 315 MHz dominates car key systems in the U.S.
- 433 MHz is much more common in Europe.

Why 915 MHz Won for IoT

Compared to 315 MHz:

- Higher allowed power (in ISM band)
- Better standardized ecosystem
- LoRaWAN support
- Industrial adoption
- Better module availability

So for student IoT projects:

-> 915 MHz is almost always the better choice.

2.4 GHz — Global Band

- Works worldwide
- Wi-Fi & Bluetooth
- Easy development
- More interference
- Shorter range than 915 MHz

Best for:

- Classroom demos
- Consumer devices
- Internet-connected projects

FCC Part 15 Basics

To use ISM legally in the U.S.:

- Stay within allowed frequency range
- Respect power limits
- Follow duty cycle rules (if applicable)
- Accept interference from others

You cannot:

- Cause harmful interference